ME101 SOME THEORIES IN MECHANICAL ENGINEERING

Section:

Final Exam:

Word Component - How to Write a Laboratory Report

By

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1. ABSTRACT

Provide an abstract of about 100-150 words.

2. OBJECTIVE

In this experiment the motion of a freely falling body is studied using a video motion tracker software to determine the acceleration of gravity. Two objectives can be stated as:

Objective 1: Determine the g based on the calculated data by tracker and excel.

Objective 2: Perform a simple error analysis based on the uncertainty assosicated with the calculation.

3. THEORY

Gravity is a force that exists between the Earth and the objects that are near it. On Earth all bodies have a weight, or downward force of gravity, proportional to their mass, which Earth's mass exerts on them. Gravity is measured by the acceleration that it gives to freely falling objects.

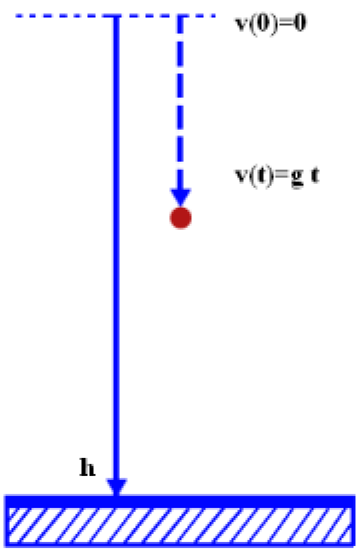

Figure 1. Depiction of the experiment

Gravitational acceleration is an expression used in physics to indicate the intensity of a gravitational field. It is expressed in meters per second squared (m/s^2). At the surface of the earth, it is about 9.81. In this experiment, the aim is to find gravitational acceleration with the help of formula where *a* is equal to *g*.

$$v(t) = v_o - gt \tag{1}$$

$$y(t) = v_o t - 1/2\, g\, t^2 + y_o \tag{2}$$

The sub index $_o$ denotes initial quantities, *g* the gravity acceleration and *t*, the time. As shown in Figure 1, vertical displacement *y* is measured from the top.

4. MATERIALS

The materials used during the experiment are

1. Balls (ping pong and tennis)
2. Digital phone camera
3. Ruler
4. Software (for processing the captured motion)

5. METHOD

The scene is prepared first. A plane surface such as a wall or the whiteboard will be used as the scene. A distance measure such as marks in the whiteboard, or an attached ruler is placed in the scene. The ball is held by a student in front the scaled board. A video camera is held parallel to the scene and recording is started. The ball is released by the other student. The experiment is repeated three times. This procedure is repeated for two balls (tennis and ping pong). Videos are loaded to the tracker software. Various graphs are visualized in the Tracker Software such as the position, velocity and acceleration with respect to time. Error calculations for both experiment are performed.

6. DATA RECORDS

One experiment was performed for each tennis ball and ping pong ball. The data was loaded to the Tracker software. The velocity versus time graphs of the tennis (Figure 2) and ping pong (Figure 3) balls from the motion capture data was further processed to reveal the acceleration.

7. DATA ANALYSIS

The data from the motion capture software was tabulated in Tables 1 and 2. Calculations of velocity and acceleration is based on numerical derivative algorithms.

Table 1. Tennis (left) and ping pong (right) ball motion results

Tennist				Ping-pon			
(s)	y(m)	a (m/s^2)	V(m/s)	t(s)	y(m)	a (m/s^2)	V (m/s)
0.00	0.00	9.78	1.68	0.00	0.00	9.78	1.68
0.03	0.04	7.88	1.86	0.03	0.04	7.88	1.86
0.07	0.11	6.13	1.94	0.07	0.11	6.13	1.94
0.10	0.22	5.67	2.14	0.10	0.22	5.67	2.14
0.13	0.35	9.77	2.33	0.13	0.35	9.77	2.33
0.17	0.47	6.10	2.64	0.17	0.47	6.10	2.64
0.20	0.53	10.36	2.86	0.20	0.53	10.36	2.86
0.23	0.66	8.34	3.12	0.23	0.66	8.34	3.12
0.27	0.75	9.46	3.54	0.27	0.75	9.46	3.54

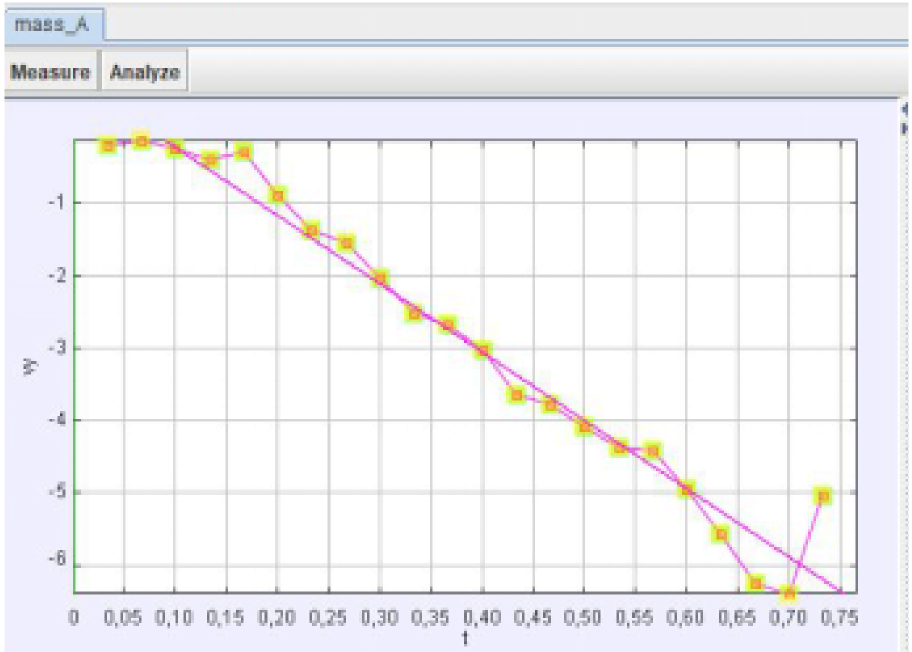

Figure 2. Tennis ball data points from the Tracker graphical user interface

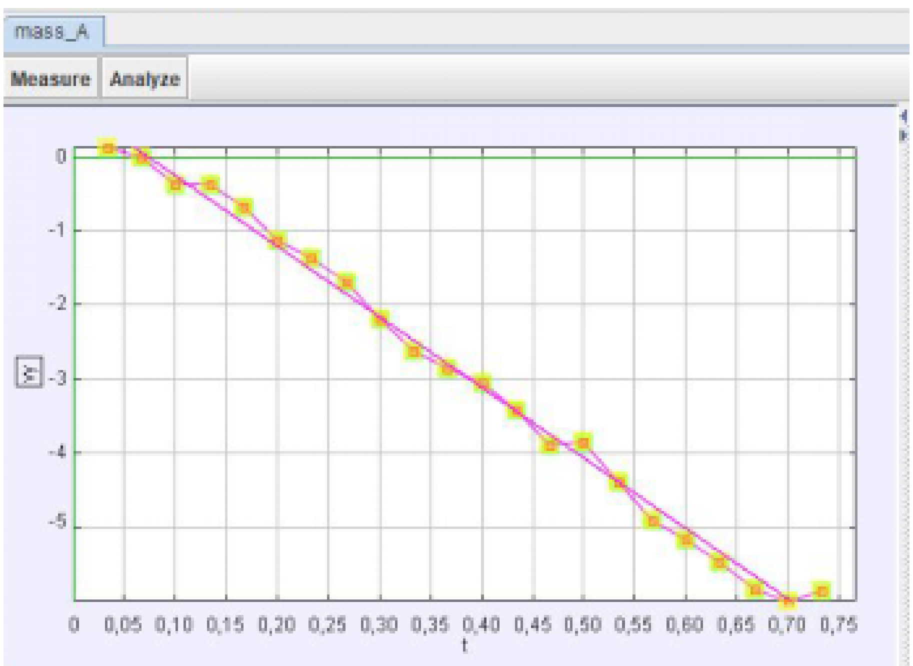

Figure 3. Ping pong ball data points from the Tracker graphical user interface

The data was further processed by passing it Excel, by a cut (Ctrl-X) and paste (Ctrl-V) operation. The data looks the same way, and slope of the temporal velocity curve represents the gravitational acceleration constant, g = 9.81 m/s^2.

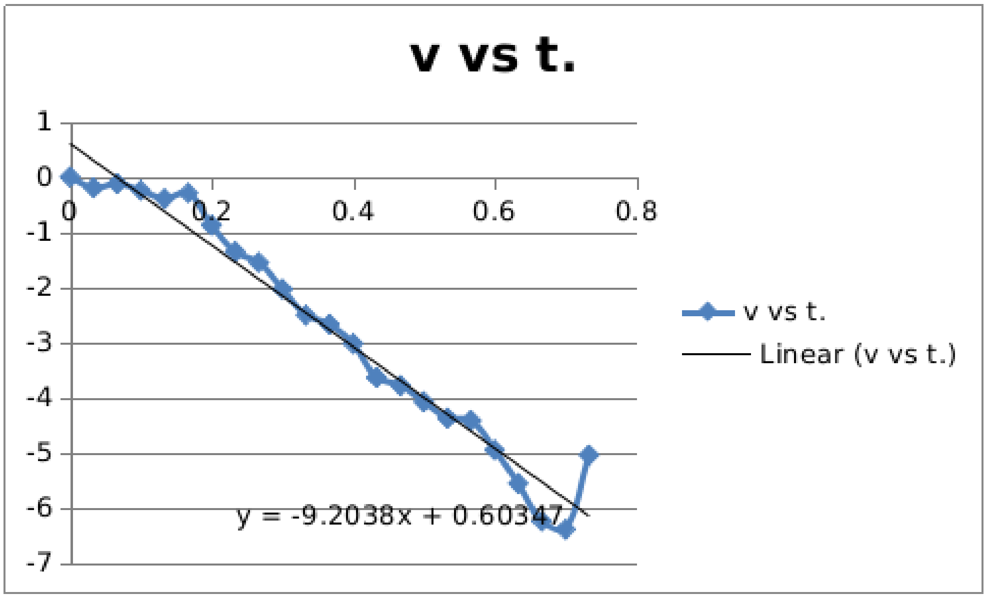

Figure 4. Plot of velocity versus time from data imported into Excel

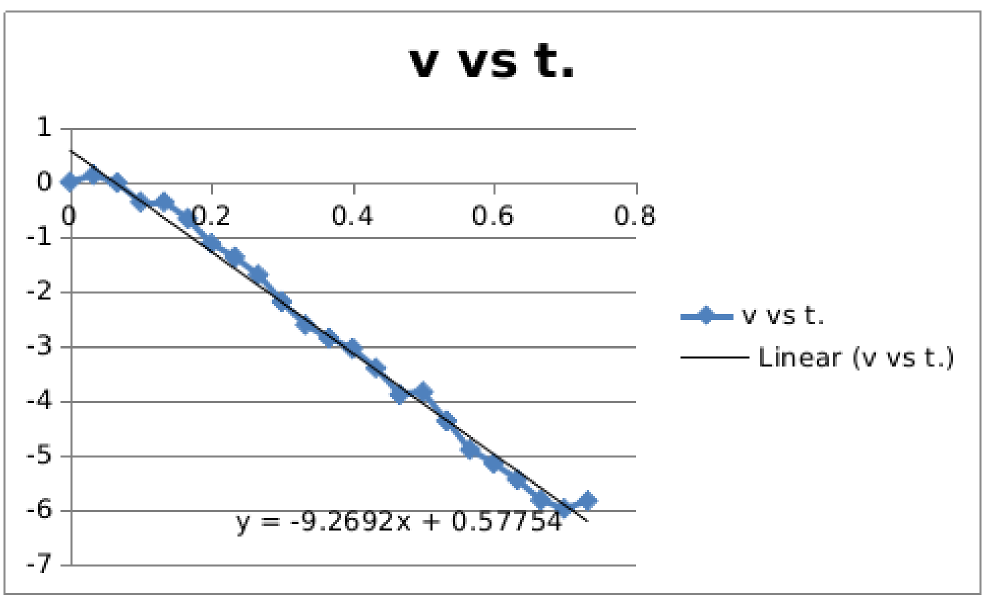

Figure 5. Plot of velocity versus time from data imported into Excel

8. DISCUSSION

An object is in free fall motion when it's motion is controlled by gravity. Objects have a downward acceleration.

The aim of this experiment was to determine the value of gravitational constant, g= 9.81 m/s^2. For this purpose the Tracker Software was used to process the images captured in the laboratory project workshop. Then, the calculations were validated by exporting them to Microsoft Excel.

The independent experiments on two different balls showed that, if the measurements can be made accurate enough, it can be shown that the gravitational constant is independent of the objects' mass.

The mass of the tennis ball was between 56.70 – 58.47 grams and the mass of the ping-pong ball was 2.7 grams. The accuracy of acceleration calculations were in error from the 9.81 m/s^2 by almost +-9%, we consider this close enough for educational purposes.

There are various reasons for the level of accuracy obtained. One of the reasons was seen as human mistakes, while marking the scene, using the camera. The depth of the length marker and the plane of motion was not the same. There will be an error due to this also. Finally, the accuracy of results may also be affected by air friction.

The gravitational acceleration value from Tracker Software is not same with the value from Microsoft Excel. The difference may be due to each program using its own numerical algorithms for differentiation of data.

9. CONCLUSION

An object is in free fall motion when it's motion is controlled by gravity. Objects have a downward acceleration.

REFERENCES

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